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Price Targets



4.1 Classical Chart Patterns

Price Targets

Richard W. Schabacker, in his book *Technical Analysis and Stock Market Profits*, discusses a very important framework for trading classical chart patterns:

The trader in stocks (any instrument in today's world) has three questions to answer before taking a position. They are:

1. When has the price move started?
2. In which direction should prices move, up or down?
3. How far can the movement carry before it is reversed, or halted into an unprofitable congestion?

The purpose of this unit is going to be to answer these three questions.

Before we start, know that both Edwards and Magee as well as Richard W. Schabacker discuss price targets as *guidelines* and not as *rules*. In fact, Schabacker writes, "It is our conclusion based on a critical examination of many hundreds of examples that no reliance can be placed in any of the theories that have been suggested,"¹ when referring to price target calculations derived from classical chart patterns.

Reversal Patterns

I have always been an advocate of using semi-log scale for charting. One of the reasons is that breakdowns and breakouts will take place much earlier than on a linear scale. Another reason is that semi-log scale allows for better long-term analysis of trends. All of the charts in this units are semi-log scale. This information will be important in measuring price targets.

Edwards and Magee, in their book *Technical Analysis of Stock Trends*, discussed price targets for both flags and pennants in uptrends and downtrends. Their conclusion is interesting, and can be applied to other chart patterns as well. They found that "...[a]dvances from Flags or Pennants in uptrends generally go farther (in terms of points or dollars) than the preceding move, whereas declines may not carry quite so far. Hence, the formula is best applied on a semi-logarithmic chart by measuring actual chart distance rather than by counting points."²

Thus, a logical way to measure price objectives to the downside is to use percentage changes and not points or dollars. This will generate conservative price targets. With upward breaks, taking point or dollar changes will generate conservative price objectives.

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Unit

4.1 Classical Chart Patterns

chart pattern at its widest point. This means the percentage distance from the highest level of the head to the neckline. The number is then subtracted from the neckline to find the price objective.



FIGURE 4.1.1 Weekly candlestick chart with volume histogram and head and shoulders top

Source: Metastock

In Figure 4.1.1, Hilong formed two different H&S top reversal patterns. First, the smaller pattern marked with the smaller green letters will be discussed. Figure 4.1.1 shows the depth of the short-term H&S top from the head to the neckline was equivalent to roughly 22% ($(\$5.4 - \$6.9) / \$6.9 = 21.4$). With the breakdown, this number was extrapolated lower to offer a price objective of \$4.2, a 22% drop from the neckline ($\$5.4 \times .22 = \1.2 and $\$5.4 - \$1.2 = \$4.2$). The price target was met in the week following the breakdown.



FIGURE 4.1.2 Weekly candlestick chart with volume histogram and head and shoulders top

Source: Metastock

The larger scale H&S top, marked with larger blue letters, is a 15-month-long pattern featured in Figure 4.1.2. The price decline from the highest point of the head to the neckline was measured as 46% ($(\$3.7 - \$6.9) / \$6.9$). With the breakdown of the neckline at \$3.70, 46% was subtracted from the neckline, resulting in a price target of \$2.0 ($3.7 \times .46 = 1.7$ and $3.7 - \$1.702 = \2). The price target was met during the downtrend that followed the breakdown of the neckline.

H&S Bottom

the difference between the neckline and the lowest point of the head is added to the neckline where the breakout takes place. The only difference, as mentioned before, is the measurement of the distance. With bullish setups, upward price targets are measured in points or dollars rather than percentage changes. To the upside, absolute price change will result in a conservative price target.



FIGURE 4.1.3 Daily candlestick chart with volume histogram and head and shoulders bottom

Source: Metastock

Figure 4.1. 3 shows the Saudi Arabia Mining Company formed a four-month-long H&S bottom reversal with the neckline at \$40.5. The distance between the lowest point of the head and the neckline was \$5 ($\$40.5 - \35.5). When added to the neckline, we found \$45.5 as the price objective ($\$40.5 + \5). Following the breakout in December 2023, price quickly rallied to the chart pattern price target at 45.5 levels.

Symmetrical Triangle

Edwards and Magee, in their book *Technical Analysis of Stock Trends*, discussed the measuring rule for the symmetrical triangle slightly different than what the community of classical chartists have been applying. According to Edwards and Magee, when dealing with an upward movement (upside breakout), draw from the top of the first rally that initiated the pattern (in other words, from its upper-left corner) a line to the bottom boundary. This line will slope up and away from the pattern to the right. Prices may be expected to climb until they reach this line. The same rule applies to a descending move.

It is important to note that the technique Edwards and Magee applied in measuring price objectives for the symmetrical triangle is again conservative in comparison to the targets of the new generation of classical chartists. Both methods will now be illustrated.



FIGURE 4.1.4 Weekly candlestick chart with volume histogram and conservative price target

Source: Metastock

Figure 4.1.4 shows the conservative method of Edwards and Magee. The top of the first rally, which initiated the pattern, was at \$6. The breakout of the symmetrical triangle took place at \$5.6. The price target of \$6 was met in two weeks.



FIGURE 4.1.5 Weekly candlestick chart with volume histogram and modern price target

Source: Metastock

The approach taken by the new generation of classical chartists is shown in Figure 4.1.5. Take the depth of the pattern in absolute price terms (with upward breaks) and add that to the breakout level above the upper boundary of the symmetrical triangle. In the case of China United Network Communications Ltd., the depth of the pattern at its widest point equaled \$1.5 (\$5.6 – \$4.1). Added to the breakout level, we forecasted a price objective of \$7.1 (5.6 + \$1.5 = \$7.1). This target is much higher than the Edwards and Magee method.

Ascending and Descending Triangle

Figure 4.1.6 is a weekly chart of Jardine Cycle & Carriage Ltd. Price formed a two-year-long ascending triangle that acted as a bottom reversal. The depth of the pattern at its widest point measured \$7.1 (\$23.9 – \$16.8). At the time of the breakout above the horizontal boundary at \$23.9, the difference of \$7.1 was added to find the possible chart pattern price target of \$23.9 + \$7.1 = \$31. The price target of \$31 was met several months after the breakout.



FIGURE 4.1.6 Weekly candlestick chart and volume histogram with ascending triangle

Source: Metastock

Figure 4.1.7 shows a weekly chart of Delivery Hero. Price formed an 11-month-long descending triangle that acted as an uptrend reversal pattern following a three-year uptrend. The depth of the pattern at its widest point measured as 31% ($(\$100 - \$145) / \$145$). To calculate the price objective, 31% is subtracted from the breakdown level ($\$100 \times .31 = \31 , and $\$100 - \$31 = \$69$). Price reached \$69 and then continued to fall even lower. Again, it is important to note that price targets should be used as guidelines only. Also, by taking the percentage change for bearish setups, we are aiming for conservative price targets.



Figure 4.1.7 Weekly candlestick chart with volume histogram and descending triangle

Source: Metastock

Rectangle Top (Double and Triple Top)

Richard W. Schabacker mentions that the larger and clearer the pattern, whether it be a reversal or a continuation formation, the more extensive the movement of prices we may expect to follow its completion.

This is probably one of the most important pieces of information that we need to keep in mind when measuring price objectives. It is also the main reason why we should always accept price targets as guidelines rather than rules.

In Figure 4.1.8, HarborOne Bancorp formed a 17-month-long rectangle consolidation, a multiple-top pattern that acted as a top reversal. With tight and lengthy chart patterns like these, one can expect price to extend two to three times the initial price target calculation.

\$11.5 our price target ($\$12.5 \times .15 = \1.7 , and $\$12.5 - \$1.7 = \$11$). Given that this is a tight and lengthy consolidation with a bearish resolution, where price can fall with its own gravity, we can comfortably expect two to three times the initial price targets to be reached.

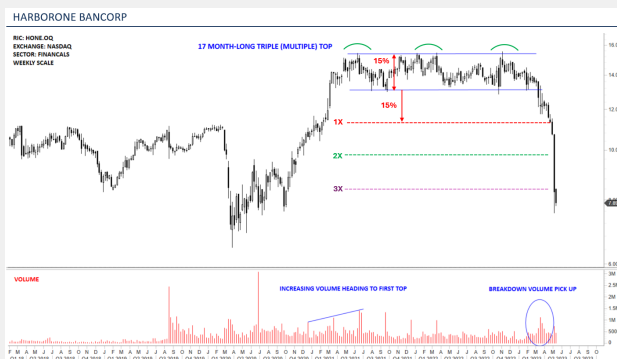


FIGURE 4.1.8 Weekly candlestick chart with volume histogram and a triple top

Source: Metastock

Continuation Patterns

Rectangle Continuation

Figure 4.1.9 is another example of a lengthy and tight chart pattern, where breakouts might achieve price targets that are two to three times their initial targets. The chart is monthly candlesticks for the Türkiye BIST Bank Index between 1999 and 2024. The index entered a lengthy sideways consolidation in 2012, and it lasted an entire decade. During that period, the index remained in a tight range and formed a rectangle chart pattern. It ultimately acted as a bullish continuation pattern upon the breakout in 2022. Given that this was such a long and tight consolidation, one can easily expect the uptrend to extend beyond the initial price objective, which was calculated as 2,760, 880 points above the well-defined horizontal boundary of 1,880 and 1,000. The index far exceeded three times its price target.

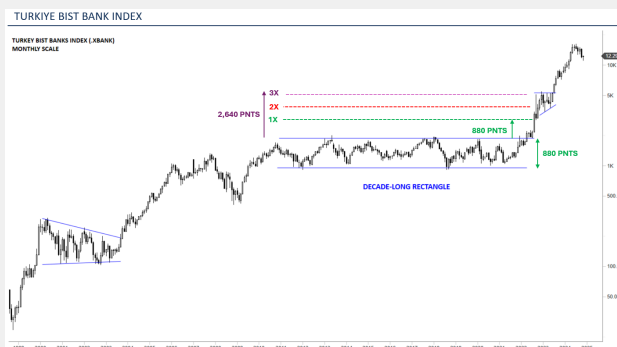


FIGURE 4.1.9 Monthly candlestick chart and a rectangular continuation pattern

Source: Metastock

Flags and Pennants

pennant as to the flag. They are both patterns that form after a steady and steep price movement.

To generate price targets, calculate the distance between where the initial move for the flag or pennant pattern started (note that the first measurement is taken from the level where price leaves the previous congestion up to the peak of the flag or pennant) and the minor reversal level where the pattern started forming. Then, measure the same distance from the point where price breaks out of the flag or pennant in the same direction. The level calculated is usually the minimum expectation for this type of short-term continuation pattern.



FIGURE 4.1.10 Daily candlestick chart and volume histogram with a bull flag

Source: Metastock

Figure 4.1.10 is a daily chart of Axalta Coating Systems. Price broke out of its downtrend with a breakaway gap. The minor high, labeled with a green dashed horizontal line, is taken as the breakout level and the starting point for first move. The advance to the point where the first minor reversal took place in the flag pattern equaled \$3.11 (\$31.11 – \$28). Price objective is calculated by adding the initial difference to the breakout level of the short-term bullish consolidation (\$30.8 + \$3.11 = \$33.91).

¹ Schabacker, Richard. *Technical Analysis and Stock Market Profits*, Harriman Definitive Edition. Hampshire, U.K.: Harriman House, 2021.

² Edwards, Robert, John Magee, and H.W.C. Bassetti. *Technical Analysis of Stock Trends*, 11th edition. New York, NY: CRC Press, 2018.

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Setup, Signal, and Reward-to-Risk Ratio

4.1 Classical Chart Patterns

Setup, Signal, and Reward-to-Risk Ratio

So far, by acquiring the knowledge of how price targets are generated, we have tackled Schabacker's third point: How far can the movement carry before it is reversed, or halted in a long and unprofitable congestion?

We will now focus on Schabacker's first and second points regarding what constitutes a signal, and how we know if the breakout or breakdown is confirmed.

In almost every charting book, one reads that price needs to close above or below the pattern boundary by a decisive margin to confirm a breakout or breakdown. What is that decisive margin? How do we know when price action is enough to call it a breakout or breakdown, and do we also need to see an increase in volume for confirmation?

Welcome to the "craft" of classical charting and technical analysis. This is not a science, and maybe not an art. It is more likely a craft, where the practitioner starts applying the tools and techniques they are comfortable with.

Edwards and Magee suggested the trigger for a breakout to be a 3% daily close above or below the pattern boundary to confirm a breakout or breakdown. This was their scientific approach for deciding if a breakout or breakdown had taken place. This method, like all methods, has its drawbacks. One issue with using the 3% method arises with higher-priced stocks. A 3% breakout confirmation can result in acting too late following a breakout. With a \$600 stock, a 3% breakout confirmation would not occur until price reached \$618. Using low-priced instruments, such as currencies or futures, a 3% breakdown confirmation might be too early for decision making. With a \$1 instrument, a 3% breakdown would occur when price reached \$0.97.

Another method for confirming breakouts and breakdowns, rather than using a fixed percentage or dollar amount, is adjusting for the instrument's daily volatility. Average True Range (ATR) is a good measurement for the instrument's volatility. Taking a multiple of its ATR and waiting for that amount of movement past the pattern boundary can offer an objective method of decision making that adjusts for each instrument's own volatility.

Here are some examples using both methods, comparing the rewards to risks and exploring ways to maximize the reward-to-risk ratio with breakout or breakdown setups.

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Unit

4.1 Classical Chart Patterns



FIGURE 4.1.11 Daily candlestick chart with a 200-period moving average and a head and shoulders bottom
Source: Metastock

Figure 4.1.11 shows a daily chart of Franco-Nevada Corp. between March 2022 and November 2022. The stock is listed on the Toronto Stock Exchange. Price formed a four-month-long H&S bottom reversal pattern with the neckline at \$172.8. According to Edwards and Magee, the stock needs to close above the chart pattern boundary by 3%, or at \$177.98. As shown on the chart, that would make the entry late and require a close well above the pattern's boundary.

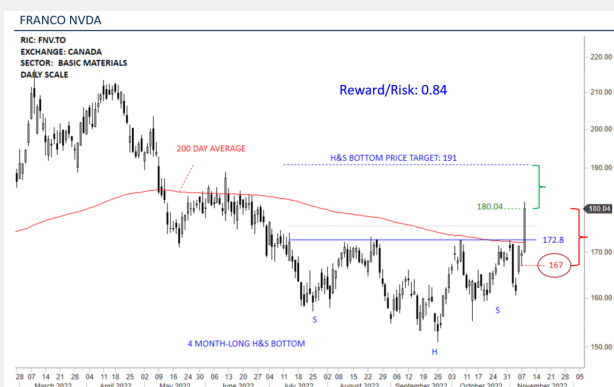


FIGURE 4.1.12 Daily candlestick chart with a 200-period moving average and a head and shoulders bottom
Source: Metastock

Figure 4.1.12 shows the ideal place to place a protective stop-loss is below the pattern's boundary after a breakout takes place. If we take the low of the previous day's candle as a stop-loss, the reward-to-risk ratio, reward being the price difference between the price objective and the daily close that confirmed the breakout and risk being the difference between the daily close that confirmed the breakout and the stop-loss, is equivalent to 0.84. Reward: $\$191 - \$180.04 = \$10.96$. Risk: $\$180.04 - \$167 = \$13.04$. Reward-to-risk ratio: $\$10.96 / \$13.04 = .84$. This means that the potential profit on this trade is 84% of the potential loss, or that the possible profit is less than the possible risk. This clearly is not a lucrative or safe trade for a breakout trader.



FIGURE 4.1.13 Daily candlestick chart with a 200-period moving average and a head and shoulders bottom
Source: Metastock

Given that chart pattern and price target cannot be changed, we can improve the reward-to-risk ratio by either acting on the breakout earlier, applying a tighter stop-loss, or both. Figure 4.1.13 takes into account that a trader or investor will not wait for a daily close, but act on the breakout as it takes place using a buy-stop order placed at the Edwards and Magee 3% breakout confirmation level of \$178. This order will buy shares when price reaches \$178. Also, we have adjusted the protective stop-loss order to the close of the previous day to \$169.3 from \$167. With this slight adjustment, we are able to improve the reward-to-risk ratio to 1.5, which is far better than 0.84. Reward: $\$191 - \$178 = \$13$. Risk: $\$178 - \$169.3 = \$8.7$. Reward-to-risk: $\$13 / \$8.7 = 1.5$. This means that the potential profit on this trade is 150% of the potential loss, or that the possible profit is 1.5 more than the possible risk.

Next, we can try to maximize the reward-to-risk ratio even further by using ATR to confirm the breakout instead of the 3% rule. Average True Range is a volatility indicator showing the average price variation of assets within a given lookback period. The True Range indicator is taken as the greatest of the following: current high less the current low; the absolute value of the current high less the previous close; or the absolute value of the current low less the previous close. The ATR is then a moving average of the selective period of the true ranges.

While the ATR does not tell us the direction of the breakout, it can be added to the closing price and then the trader can buy whenever the next day's price trades above that value. Many trend-following systems or breakout systems use ATR values to confirm breakouts. If a stock is making a daily move that exceeds an ATR value, it is considered significant and a breakout trading signal or confirmation.

We can incorporate ATR into classical chart pattern trading to decide on what actually constitutes a breakout signal. Given that we have seen at times that the Edwards and Magee 3% breakout guideline ended up being a late entry or exit signal, we can easily benefit from a more adaptive method that takes into account ATR values.



FIGURE 4.1.14 Daily candlestick chart with a 200-period moving average and a head and shoulders bottom
Source: Metastock

We will not get into details of ATR calculations here. We will only focus on the application of ATR as a breakout confirmation signal. In Figure 4.1.14, we are looking for a breakout confirmation level that will complete the four-month-long H&S bottom. It is possible to take a 10-period ATR and use half of the value, or a 0.5 multiple, above and below the horizontal boundary to decide on a protective stop-loss and a breakout confirmation level. Our order management is a buy-stop order where the order will get executed when price reaches \$175.2. Reward: $\$191 - \$175.2 = \$15.8$. Risk: $\$175.2 - \$170.4 = \$4.8$. Reward-to-risk: $\$15.8 / \$4.8 = 3.3$. This means that the potential profit on this trade is 300% of the potential loss, or that the possible profit is three times more than the possible risk.

By applying a multiple of the ATR for entry and stop-loss placement, we are taking an objective and adaptive approach. It is objective because this can be applied across asset classes and price levels, and adaptive because the volatility of the instrument will be reflected in our calculations.

This tactical approach has clearly improved the reward-to-risk ratio.

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Breakout Trading Vs. Rangebound Trading

4.1 Classical Chart Patterns

Breakout Trading Vs. Rangebound Trading

Classical chart patterns are consolidation periods in steady uptrends or downtrends. When a chartist tries to identify the chart pattern, they are actually trying to understand if the instrument is rangebound or trending. Several chart patterns are neutral. This means that we need to wait for the actual breakout or breakdown to decide on the direction of the next price movement.

This brings us to two different trading strategies. If identified early, rangebound price action can be useful in applying mean reversion strategies. This involves selling at resistance and buying at support, until a confirmed breakout takes place. At that point, we switch to a breakout strategy.

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Unit

4.1 Classical Chart Patterns

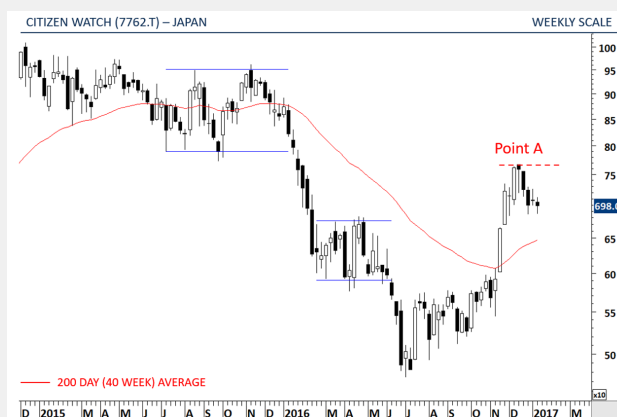


FIGURE 4.1.15 Weekly candlestick chart with a 40-period simple moving average

Source: Metastock

Figure 4.1.15 is a weekly chart of Citizen Watch Co. from 2015 to 2017. The stock is listed on the Nikkei Stock Exchange. After a lengthy downtrend, price remounted its 40-week average in late 2016. Point A shows the first reversal after the rebound. One can start looking for a consolidation above the moving average.

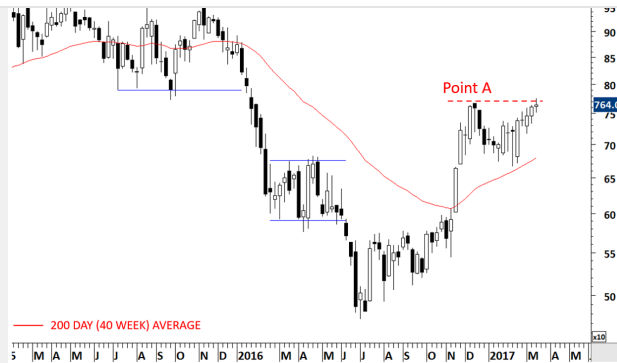


FIGURE 4.1.16 Weekly candlestick chart with a 40-period simple moving average

Source: Metastock

Figure 4.1.16 is roughly nine weeks later, and at this time the trader is wondering: “Can point A act as resistance and offer an area to initiate the reversion to the mean strategy?” At this point, we can look to the candlestick patterns for help and look for reversal candlestick patterns around potential resistance and later around potential support.

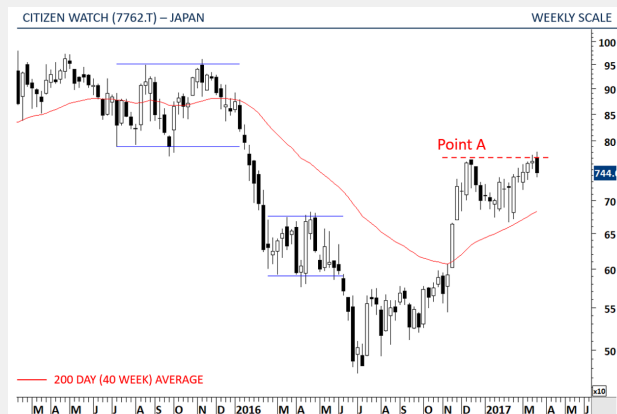


FIGURE 4.1.17 Weekly candlestick chart with a 40-period simple moving average

Source: Metastock

Figure 4.1.17 shows a bearish engulfing candlestick around potential resistance. This suggests that resistance will hold, and that the stock is likely to retreat towards the 40-week moving average and the minor low that formed in early 2017. Selling around resistance and trying to cover that short position around support would be the right rangebound trading strategy.

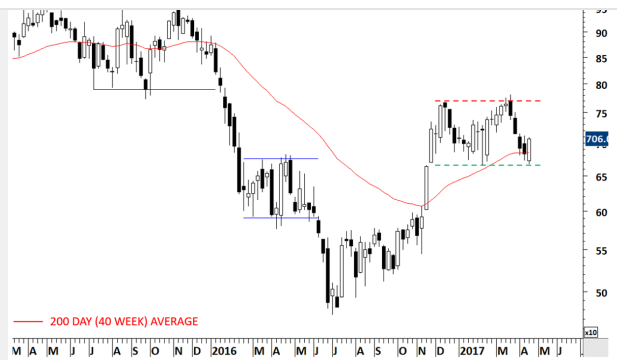


FIGURE 4.1.18 Weekly candlestick chart with a 40-period simple moving average

Source: Metastock

Figure 4.1.18 shows that following the bearish engulfing candlestick around potential resistance, price pulled back to the minor low, where it not only found support but also formed a bullish engulfing candlestick that offered a long signal. A bullish engulfing is the exact opposite of a bearish engulfing. These are outside week candlesticks formed by two candles that suggest reversals around support and resistance levels. After two tests of the upper and lower boundaries, the consolidation range became better defined. An ideal strategy is to cover the short position with the formation of the bullish engulfing at support, and then to initiate a new long position with the expectation of testing the upper boundary once again.

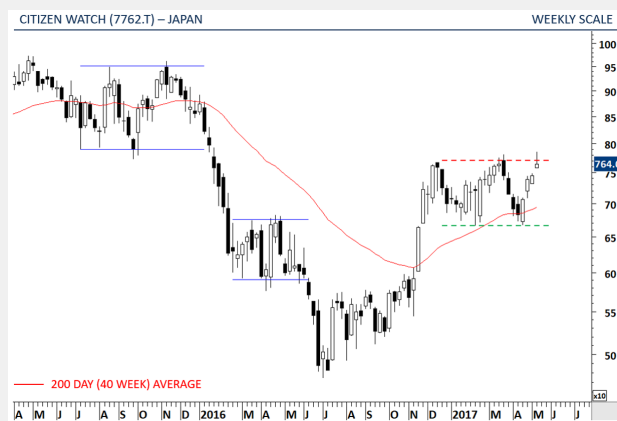


FIGURE 4.1.19 Weekly candlestick chart with a 40-period simple moving average

Source: Metastock

Figure 4.1.19 shows price returned to resistance, and given that we know the stock reversed earlier twice already, it is wise to expect some overhead resistance and supply coming to the market. Resistance is an area to exit long positions and monitor for a possible breakout (which will require the implementation of a breakout strategy) or some consolidation with price potentially returning to its average of the bottom of the range. Price printed a potential shooting star reversal candlestick at resistance.

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FIGURE 4.1.20 Weekly candlestick chart with a 40-period simple moving average

Source: Metastock

Figure 4.1.20 shows what happened after the shooting star bearish candle. Price did not retrace all the way back to the lower boundary. The 40-period moving average had been moving higher and acted as support. At this point, the pattern can be clearly identified as a rectangle. Breakout from the rectangle can be traded using a breakout strategy. You can also see that the frequency of tests at the resistance increased, showing impatience of buyers after each retreat.

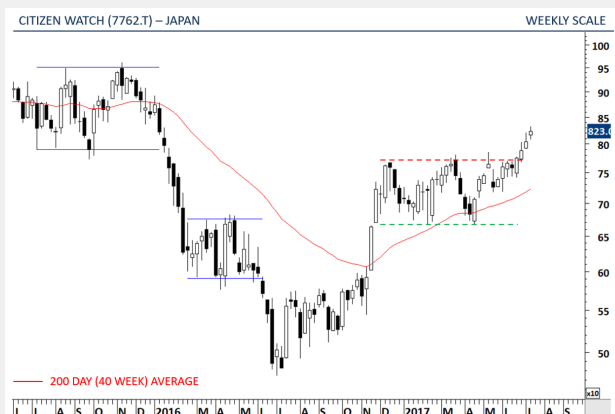


FIGURE 4.1.21 Weekly candlestick chart with a 40-period simple moving average

Source: Metastock

Figure 4.1.21 shows the eventual breakout. This completed the rectangular chart pattern and turned it into a bullish continuation pattern. Breaching of the upper boundary stressed the importance of quickly applying a breakout trading approach by buying at the time of the breakout confirmation.

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